

Enchanted Touch: Delivering Vibrotactile and Thermal Feedback Without Fingerpad Contact

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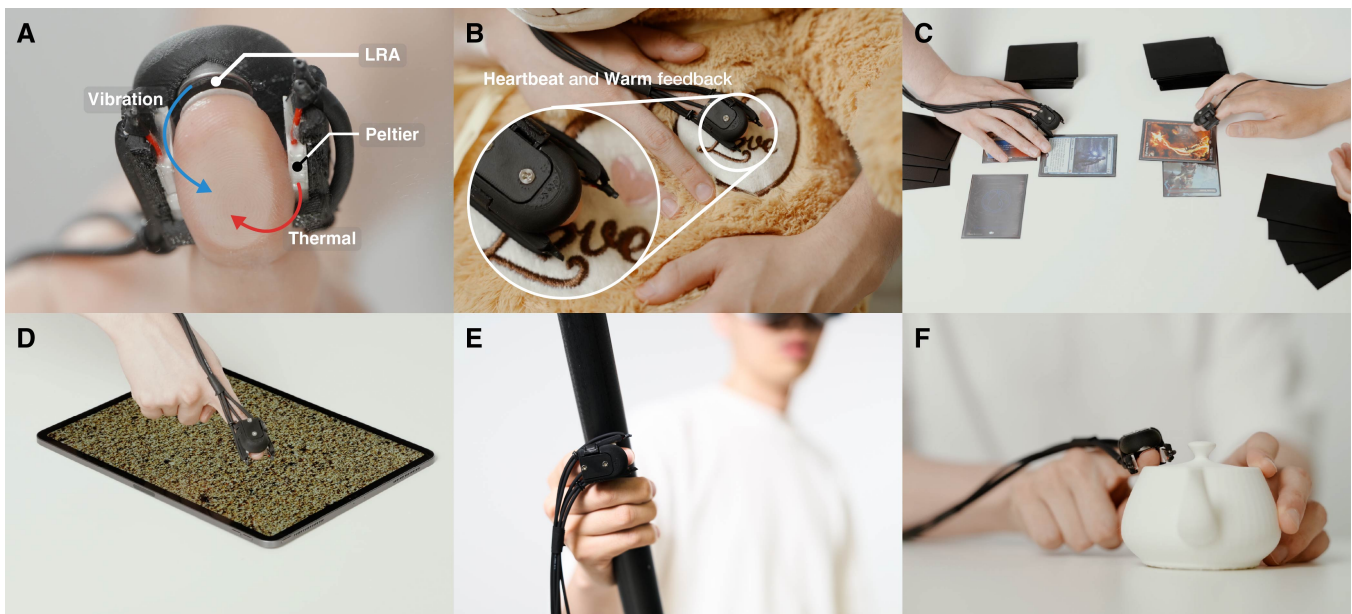


Figure 1: Enchanted Touch and its applications. (A) Device overview and fingerpad-free principle; (B) animacy for a plush toy; (C) elemental feedback on a trading card; (D) texture rendering on a tablet; (E) MR hammer striking material blocks; (F) MR texture overlay on a 3D-printed object.

Abstract

We present Enchanted Touch, a finger-mounted device that layers thermal and vibrotactile feedback onto everyday objects without covering the fingerpad or dulling textures. Conventional feel-through, on-demand, and relocated approaches require a trade-off between virtual haptic fidelity and real-world interaction. Enchanted Touch improves upon the relocated actuators by combining

two haptic illusions, redirecting both thermal sensation and vibration to the fingerpad. The prototype integrates four Peltier elements along the finger sides and one vibrator on the nail, enabling real-time presentation of warmth, coolness, and vibration while keeping the fingerpad fully exposed. Demonstrations show the device adding animacy to plush toys, delivering elemental feedback when a trading card is activated, and rendering material-specific sensations on a tablet surface. These examples show how Enchanted Touch preserves dexterity while providing high-fidelity cues, pointing toward any-object haptic interaction.

CCS Concepts

• Human-centered computing → Haptic devices.

Keywords

haptics, fingerpad free, thermal referral, nail-mounted vibration

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1 Introduction

Daily life is rich in haptic experiences originating from a wide variety of objects, including smartphones, game controllers, plush toys, and trading cards. These items offer inherent haptic information such as stiffness and fine surface texture, yet they rarely deliver additional sensations like changing temperature or localized vibration. If such feedback could be layered on top of the original feel without attenuation, the scope of any-object haptic interaction would be greatly extended.

Previous haptic-augmentation devices follow three patterns, and all face a trade-off between virtual haptic fidelity and real-world interaction. Feel-through actuators [Withana et al. 2018] wrap the fingerpad in thin films and deliver high fidelity virtual stimuli, but they mask fine surface textures and restrict available modalities. On-demand actuators [Teng et al. 2021] deploy only when required, yet their mechanisms can interfere with objects and provide merely moderate fidelity. Relocated actuators [Cai et al. 2025; Maeda et al. 2022] attach actuators to the nail or finger sides, preserving an unobstructed pad, but their expressive bandwidth remains limited. To date, no single approach has satisfied fidelity and real-world interaction simultaneously.

To tackle this trade-off, we extended the relocated-actuator and developed Enchanted Touch, a wearable device that mounts four Peltier elements along the finger sides and a miniature vibrator on the nail, leveraging two haptic illusions (Figure 1A). First, thermal referral mislocalizes temperature: a stimulus delivered to the finger's side is also perceived on the finger-pad that actually contacts the object [Sato 2016]. Second, a nail-mounted vibration illusion conveys high-frequency vibration from the nail to the fingerpad [Ando et al. 2002]. By driving both illusions together in real time, Enchanted Touch presents warmth, coolness, and vibration without covering the pad, thus preserving the dexterity needed for interacting with physical objects. Integrating this haptic feedback adds a haptic layer to the objects around users, creating an experience as if they were enchanted by "magic". For example, you can add animacy to a plush toy, include haptic feedback that matches the effects of a trading card, or recreate haptic sensations on a tablet's screen.

2 Implementation

Enchanted Touch consists of two types of actuators, a housing, and a control unit. The thermal actuators are compact 8.3 mm × 8.3 mm Peltier elements (TEFC1-03112, SolidPhysics), arranged in two pairs (total of four units) along the lateral sides of the finger. The vibrotactile actuator is a 10 mm diameter linear resonant actuator (LRA; Vibration Motor, Vybronic), mounted on the fingernail. All five actuators are independently controllable. Inspired by prior wearable haptic devices [Maeda et al. 2022], the housing incorporates a spring-based clamping mechanism that secures the device laterally to the fingertip, ensuring that the fingerpad remains fully exposed.

This fingerpad-free design preserves natural tactile contact with physical objects while allowing additional feedback to be layered. The control unit is composed of a microcontroller and dedicated driver circuits for each actuator. Enchanted Touch is wirelessly controlled via Bluetooth or Wi-Fi.

3 User Experiences

Enchanted Touch has been applied to five representative scenarios (Figure 1B-F): animacy augmentation of a plush toy, elemental feedback in card games, tablet-based texture rendering, material-specific hammer strikes in mixed reality (MR), and visual-haptic texture overlays on 3D-printed props such as a teapot. From this diverse set, our SIGGRAPH Asia 2025 demo highlights the three most intuitive and hands-on interactions:

(1) Animacy Augmentation of a Plush Toy (Figure 1B)

A plush bear features an embroidered heart on its chest. When attendees press the heart, Enchanted Touch delivers gentle warmth to the finger sides and a synchronized heart-beat vibe on the nail. Because the fingerpad is unobstructed, users still feel the toy's native softness and fiber texture, conveying a lifelike sense of animacy.

(2) Card-Triggered Elemental Feedback (Figure 1C)

During a tabletop game, tapping a spell card's emblem triggers effect-matched cues on the target player's finger: warm, low-frequency pulses for fire; cool, high-frequency buzzes for ice. Only the finger sides and nail are actuated, so the paper's stiffness and finish remain perceptible, enabling immersive yet unobtrusive feedback for trading- and board-game play.

(3) Tablet-Based Localized Texture Rendering (Figure 1D)

Dragging a stylus or fingertip across swatches of virtual metal, wood, sand, or slime modulates thermal and vibrotactile cues in real time. Although the finger still glides over bare glass, the added thermal and vibration patterns convey the feel of the materials while the fingerpad remains fully exposed.

These demonstrations show that Enchanted Touch preserves natural dexterity and object feel while seamlessly adding multimodal feedback, pointing toward any-object haptic interaction.

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